

# 1. Overview

## The Analysis of an online store's operations

**4.67M**

Turn Over

**1.468K**

Number of unique customers

**52.924K**

Number of transactions

**3.38M**

Profit

**20**

Number of product categories

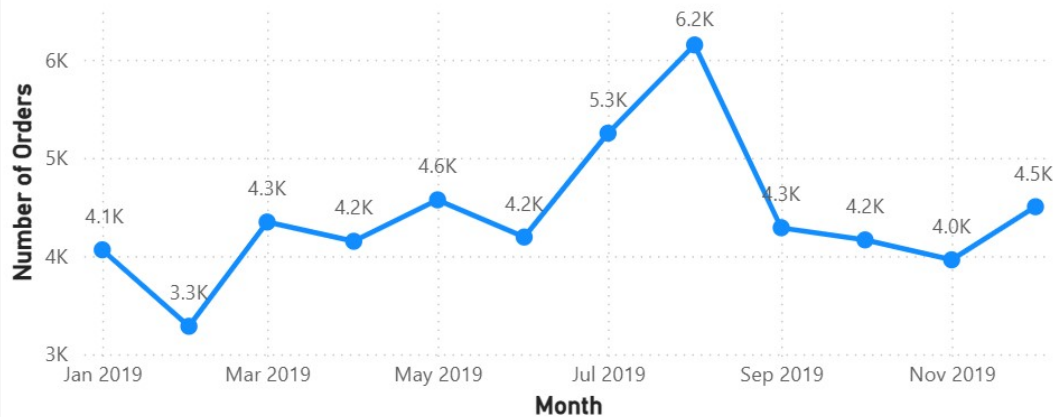
**0.67**

EBIT margin

**0.58**

Net profit margin

**Number of orders by month**

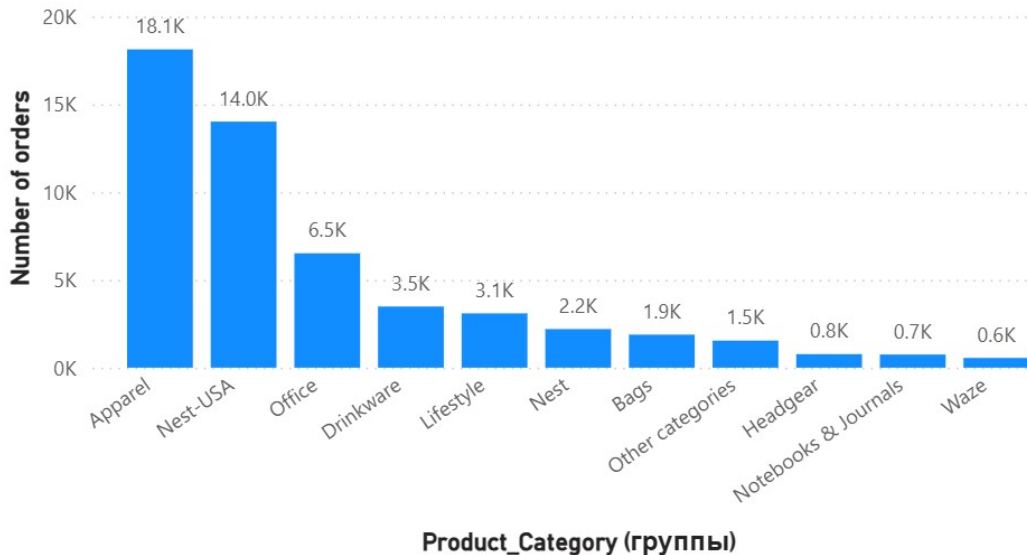


**Profit amount by month**

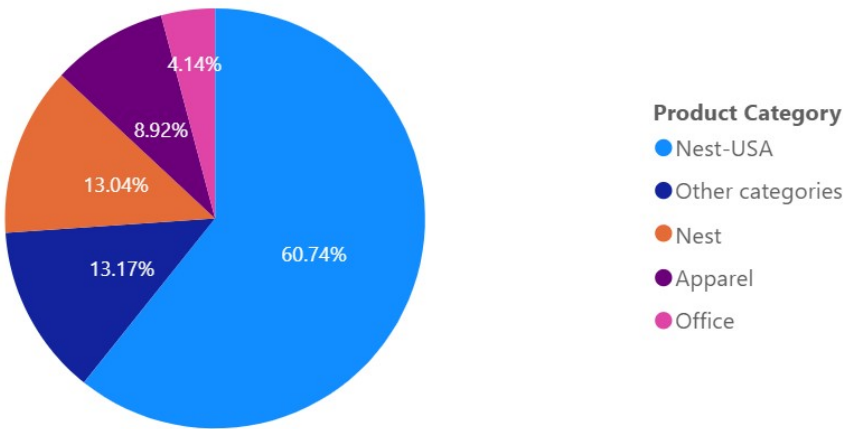


# 2. Product Categories Analysis

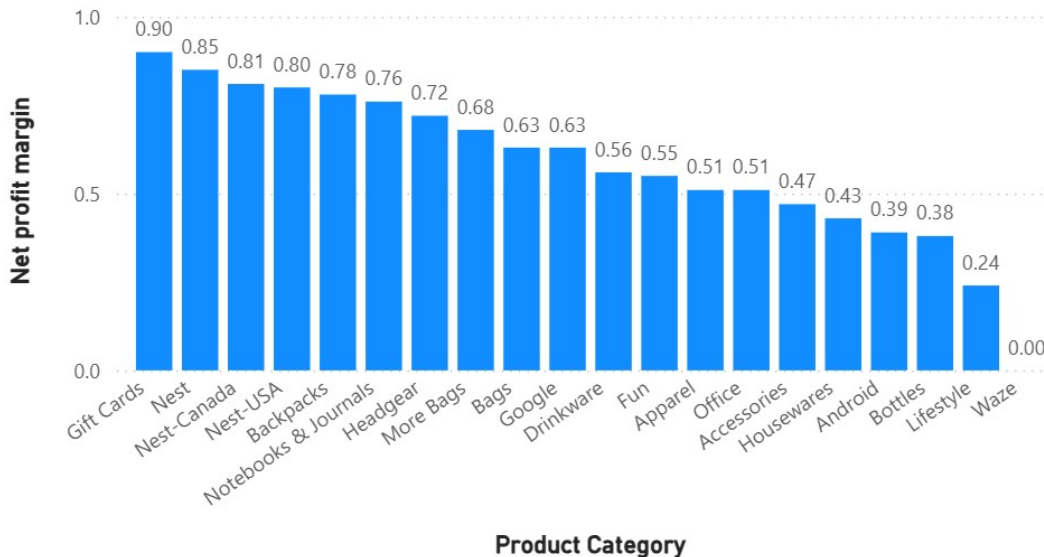
Number of orders by Product Category



Total profit by Product Category



Net profit margin by Product Category



## Executive Summary: Key Insights

### Most Popular Categories (Volume Drivers)

**Apparel** and **Nest-USA** remain the primary drivers of traffic and volume. **Apparel** leads with 18.1k orders and 1,323 customers, while **Nest-USA** closely follows with 14.0k orders and 1,260 customers.

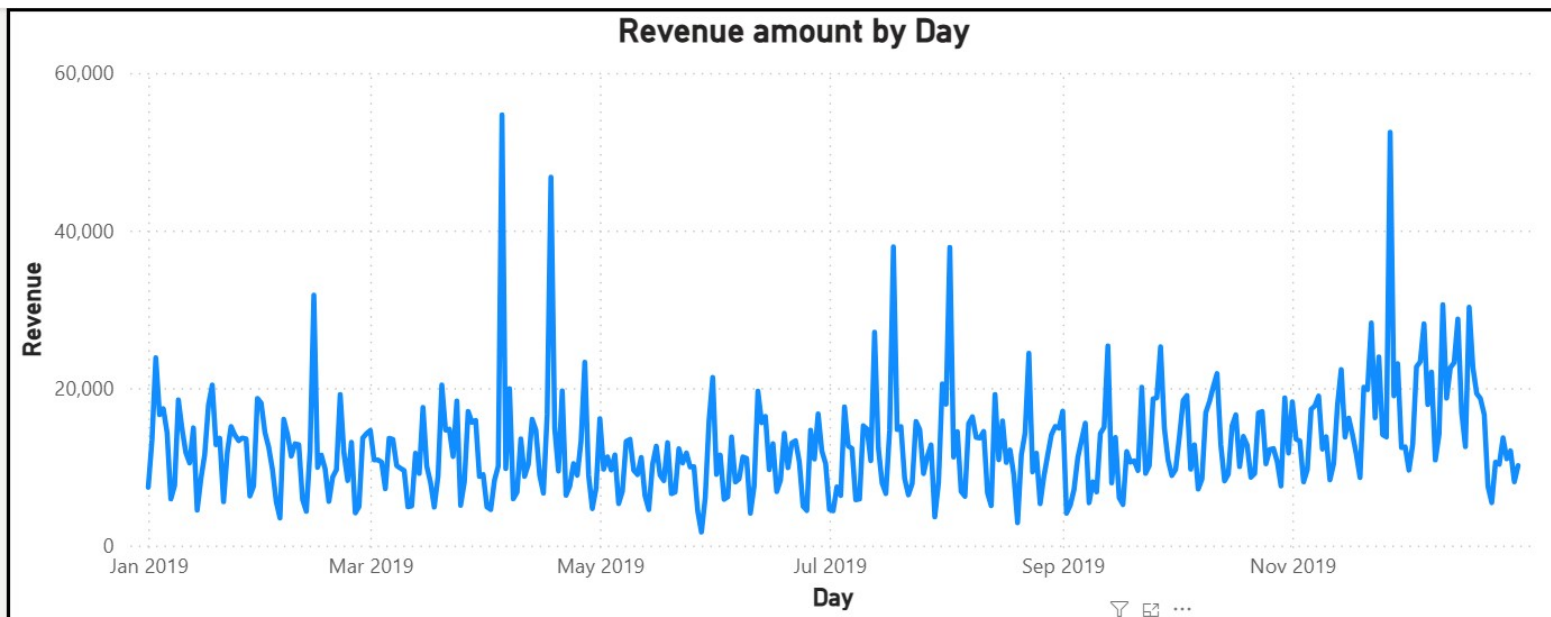
### Most Profitable & High-Margin Categories

**Nest-USA** is the ultimate financial engine, generating **60.74% of total profit** with a strong net profit margin of **0.81**. However, **Gift Cards** and **Nest** boast the highest individual profitability efficiency, leading the net profit margin chart at **0.90** and **0.85** respectively.

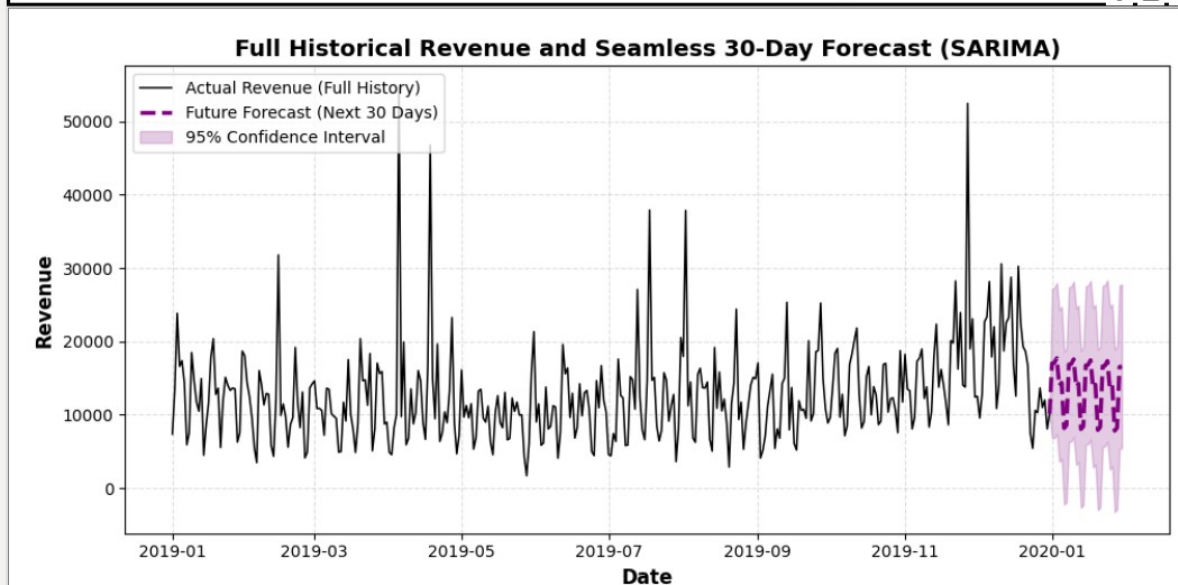
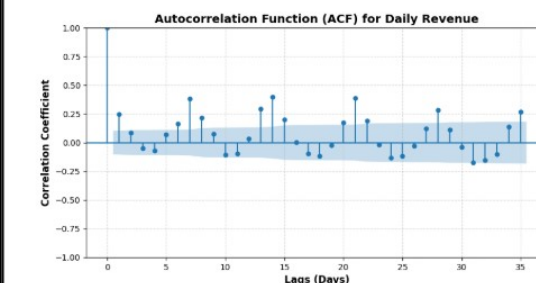
### Underperforming & Unprofitable Categories (Waze & Apparel)

**Waze** represents a critical financial leak; despite having 0.6k orders, its net profit margin drops to **0.00**, making it completely unprofitable and a candidate for strategic review. Meanwhile, **Apparel** suffers from a severe efficiency gap—despite its massive popularity, its low profit margin (**0.51**) results in it contributing just **4.14%** to total profit.

# 3. Revenue Analysis & Forecasting



| Model Name          | MAE     | MAPE (%) |
|---------------------|---------|----------|
| Seasonal Naive      | 4521.25 | 36.44    |
| ETS (A, N, A)       | 3424.85 | 28.62    |
| Autoregressive (AR) | 3939.45 | 36.82    |
| SARIMA (Best)       | 3245.87 | 27.07    |



## Executive Summary: Time Series Analysis and Forecasting Insights

### Strong Weekly Seasonality

The Autocorrelation Function (ACF) chart reveals distinct, recurring peaks at lags 7, 14, 21, and 28. This mathematically proves a highly consistent **7-day seasonal pattern** in daily revenue. Sales performance strongly depends on the day of the week, allowing for highly structured and predictable short-term demand planning.

### Model Evaluation & Accuracy Metrics

Four predictive models were trained and benchmarked against the historical daily revenue data using MAE (Mean Absolute Error) and MAPE (Mean Absolute Percentage Error):

- **SARIMA (Best):** MAE = 3,245.87 | MAPE = 27.07%
- **ETS (A, N, A):** MAE = 3,424.85 | MAPE = 28.62%
- **Autoregressive (AR):** MAE = 3,939.45 | MAPE = 36.82%
- **Seasonal Naive:** MAE = 4,521.25 | MAPE = 36.44%

### Champion Model Selection

The **SARIMA(1, 0, 1)x(1, 0, 1, 7)** model outperformed all other approaches, delivering the lowest overall error metrics. By combining both short-term autoregressive dependencies and a dedicated seasonal component, it successfully captured the rapid daily fluctuations while strictly adhering to the weekly cycles.

### 30-Day Revenue Forecast

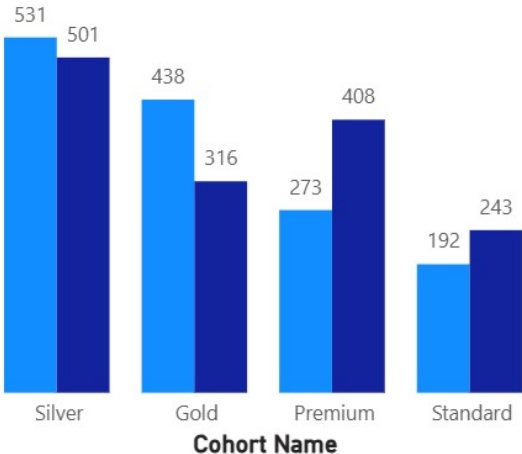
The final seamless 30-day forecast projects that the weekly wave-like revenue pattern will continue robustly into January 2020. The 95% confidence interval remains stable and bounded, indicating high model reliability and providing a solid data-driven foundation for upcoming inventory management, staffing schedules, and financial budgeting.

## 4. RFM & Cohort Analysis



## Number of customers by Cohort Name

● Kmeans Method ● Rank method



|                     | Stable_Gold | Stable_Premium | Stable_Silver | Stable_Standard |
|---------------------|-------------|----------------|---------------|-----------------|
| Number of customers | 299.00      | 486.00         | 520.00        | 129.00          |
| Average monetary    | 2,370.40    | 4,762.58       | 691.32        | 80.46           |
| Average recency     | 167.44      | 56.80          | 190.11        | 229.12          |
| Average margin      | 0.59        | 0.60           | 0.58          | 0.54            |
| Churn rate          | 0.46        | 0.16           | 0.52          | 0.63            |
| Life Time Value     | 3,041.70    | 18,462.88      | 762.90        | 69.39           |

## Life Time Value



### Key Findings: RFM & Cohort Analysis

## RFM Segment Insights

**Premium (Core Revenue Driver):** Only 486 customers generate an outstanding LTV of **18,462.88** and the highest purchase frequency (**Recency: 56 days**). They are the most valuable asset.

**Gold (Retention Risk):** Good LTV (3,041.70), but a high **Recency of 167 days** signals that this segment is actively drifting into dormancy and needs urgent re-engagement.

**Standard (Low Fit):** Underperforms critically with a **63% Churn Rate** and negligible LTV (69.39).

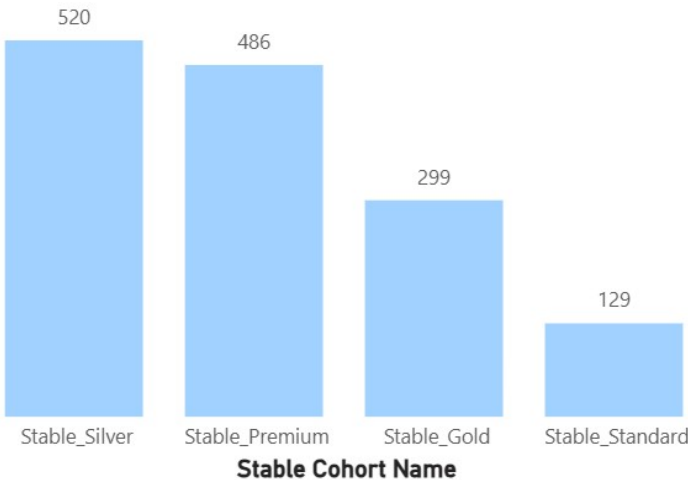
### Cohort Retention Trends

**Month 1 Drop-off:** Retention severely crashes from 100% to 6%–13% in the very first month across all cohorts, indicating a critical issue with initial customer onboarding.

**Mid-Year Resurgence:** January and February cohorts show a strong recovery in June–July (peaking at 20.5%–22.9%), which highlights a highly successful summer reactivation campaign.

**H2 Cohort Weakness:** Customers acquired from August onwards show lower engagement and significantly weaker retention trails.

### Number of customers by Stable Cohort Name

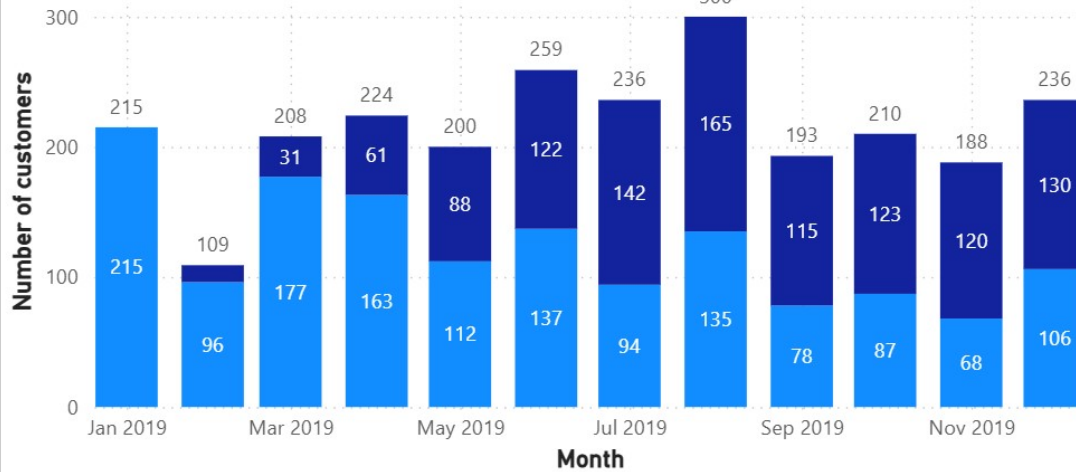
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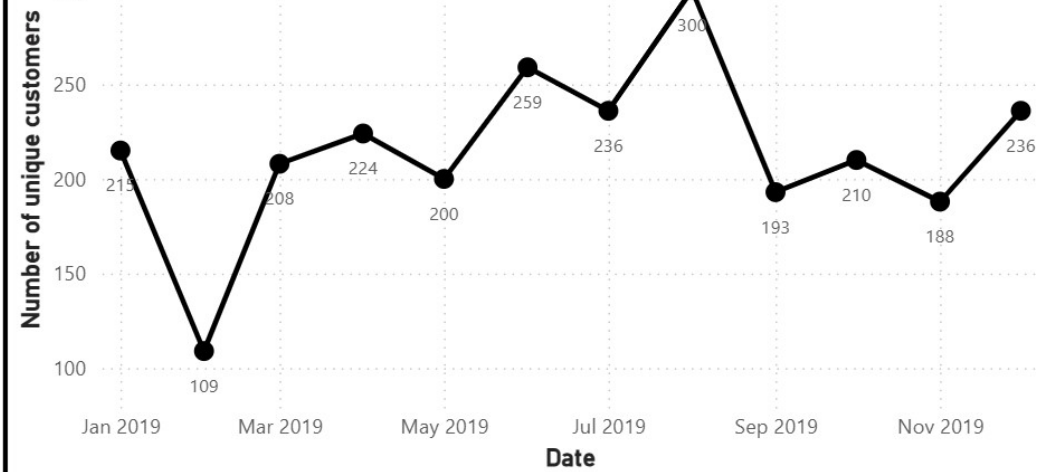
# 5. New vs Returning Customers Analysis

### Number of New and Old customers by Month

● Number of New customers ● Number of Old customers

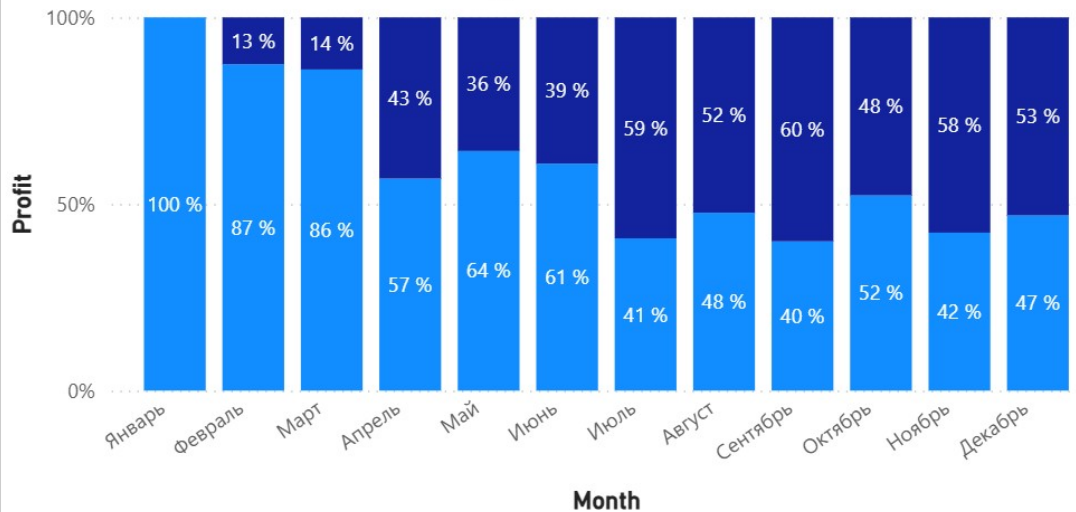


### Number of unique customers by month



### User Type Profit by Month

● New ● Old



## Executive Summary: Key Insights

### Shift in Profit Structure

At the beginning of the year (**January – March**), new customers generated the vast majority of profit (ranging from **86% to 87%**). However, starting in April, the profit share from returning customers (**Old**) began to rise steadily. In the second half of the year (particularly in July, September, and November), returning customers became the primary financial driver, contributing **up to 60% of total profit**.

### Customer Retention Efficiency

The "User Type Profit by Month" chart clearly demonstrates successful customer retention efforts. Even though the influx of unique monthly customers (top-right chart) fluctuates around 200–250, **returning customers are spending larger amounts over time**. This indicates growing brand loyalty and an increasing average order value for repeat purchases.

### Peak Performance Period (August)

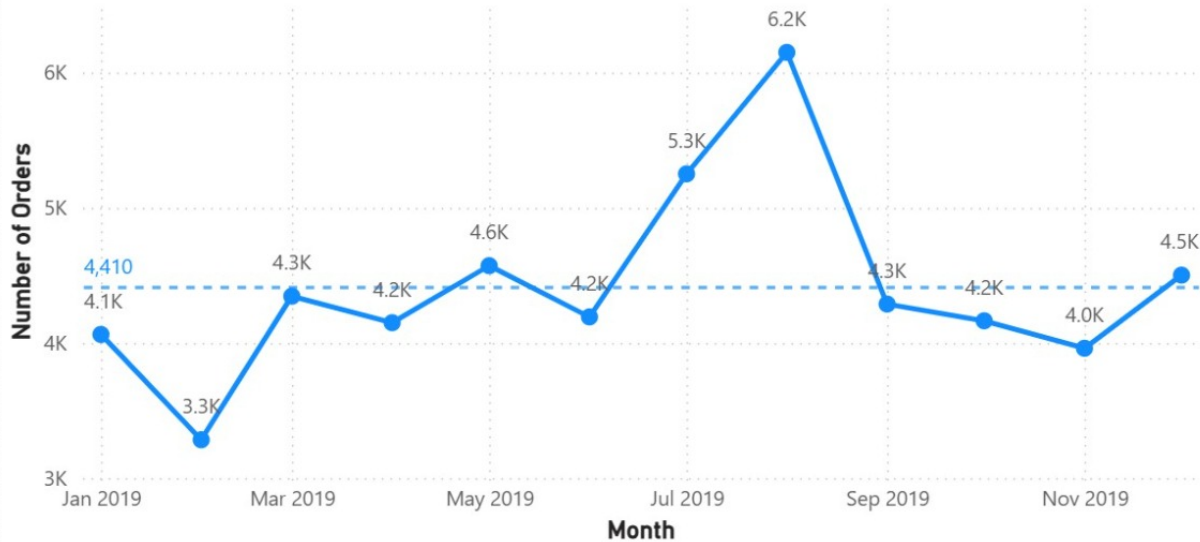
August was the peak month for customer engagement, reaching a record **300 unique customers** and the highest total number of orders. During this peak, profit was distributed almost equally: **48%** from new customers and **52%** from returning ones. This points to a highly successful late-summer marketing campaign that maximized acquisition while effectively engaging the existing base.

### Autumn Traffic Drop

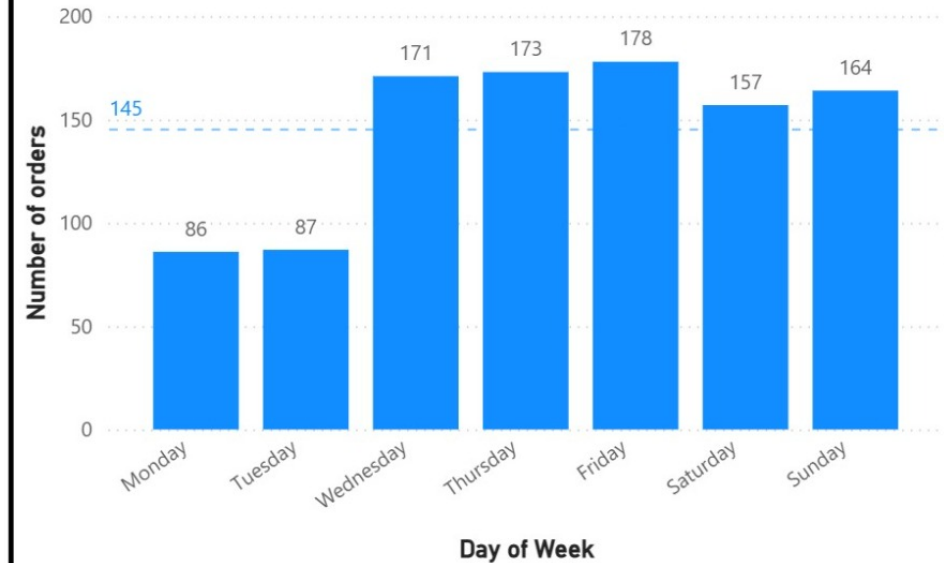
In **September**, there was a sharp drop in unique customer counts (falling from 300 to 193). Interestingly, despite this decline in overall traffic, the profit share from returning customers reached its yearly high at **60%**. This proves that during seasonal downturns, the loyal customer base protects the business from severe revenue drops.

## 6. Number of Orders Analysis

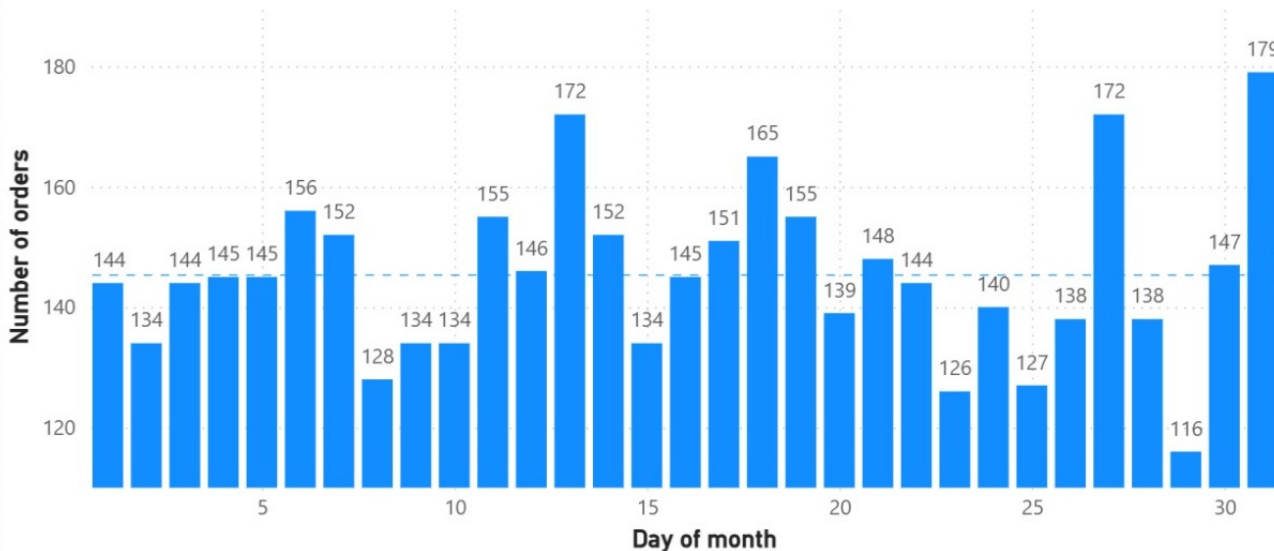
### Number of orders by month



### Number of orders by Day of Week



### Number of orders by Day of month



### Key Insights

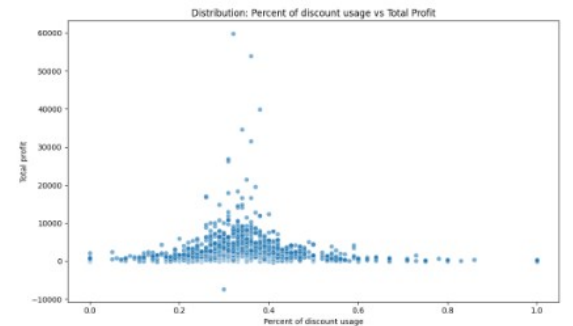
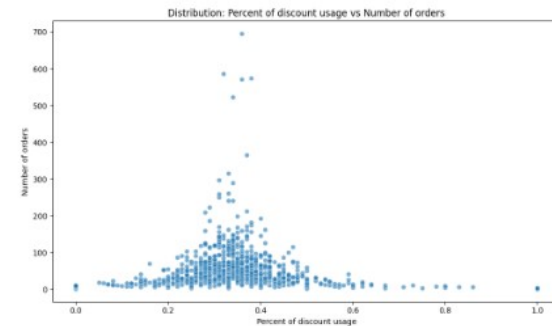
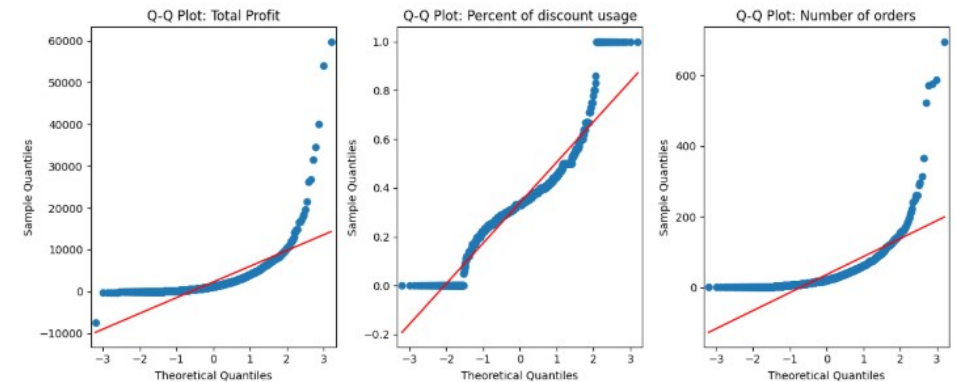
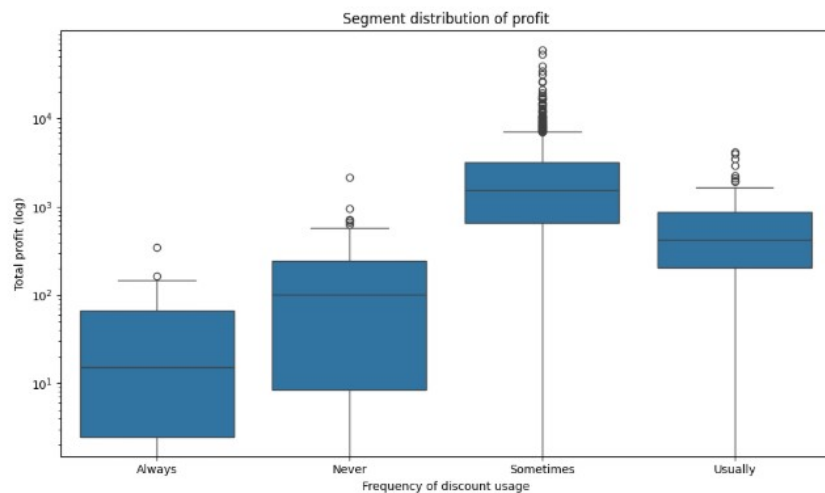
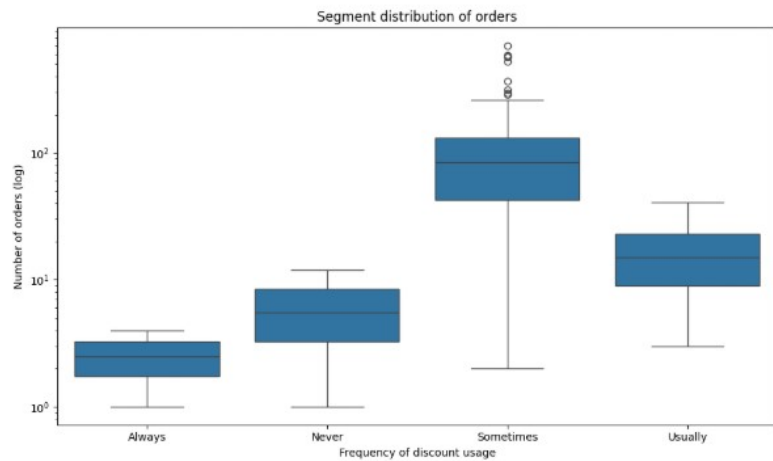
**Monthly Trend:** Order volume peaked significantly in **August 2019** at **6.2K orders**, following a steady rise from a yearly low in **February (3.3K)**.

**Day of Week Analysis:** Order activity remains highly concentrated in the middle and end of the week, with **Friday** seeing the highest volume (**178 orders**) and **Monday/Tuesday** showing the lowest engagement (under 90 orders).

**Day of Month Distribution:** Monthly orders show high volatility with notable spikes around the **12th**, **20th**, and a peak on the **31st (179 orders)**, while dropping to a monthly low on the **29th (116 orders)**.

# 7. Discount Correlation Analysis

|                                      | Always   | Never     | Sometimes    | Usually   |
|--------------------------------------|----------|-----------|--------------|-----------|
| Average profit                       | 753.54   | 7,098.07  | 79,488.01    | 5,113.73  |
| Average number of orders             | 1.32     | 2.40      | 41.07        | 11.58     |
| Average of Percent of discount usage | 1.00     | 0.00      | 0.33         | 0.62      |
| Total_profit                         | 1,081.33 | 16,640.82 | 3,302,283.22 | 64,451.50 |



## Conclusion & Statistical Insights

**Data Distribution:** Based on the Q-Q plots, none of the analyzed data series follow a normal distribution, as evidenced by the significant deviations from the theoretical reference lines.

**Correlation Limitations:** The relationship between the frequency of discount usage, the number of orders, and total profit is fundamentally non-monotonic. Therefore, applying Pearson or Spearman correlation coefficients to these variables is statistically incorrect and misleading.

**Boxplot Analysis:** Segment distribution boxplots reveal that the "Sometimes" group generates the highest total profit. In terms of order volume, the "Sometimes" and "Usually" segments heavily outperform the others, representing the primary drivers of purchase frequency.

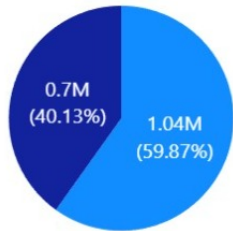
**Business Takeaway:** Total profit and order counts do not have a direct, linear dependence on how often a customer uses discounts. This indicates that discount availability is not the decisive factor driving customer purchasing behavior or overall profitability.



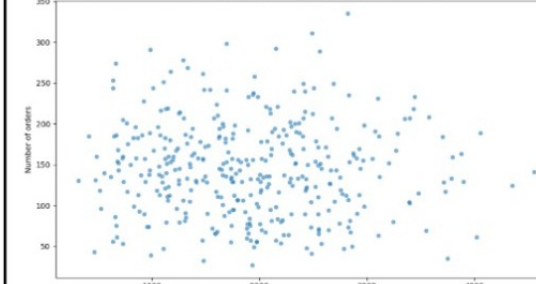
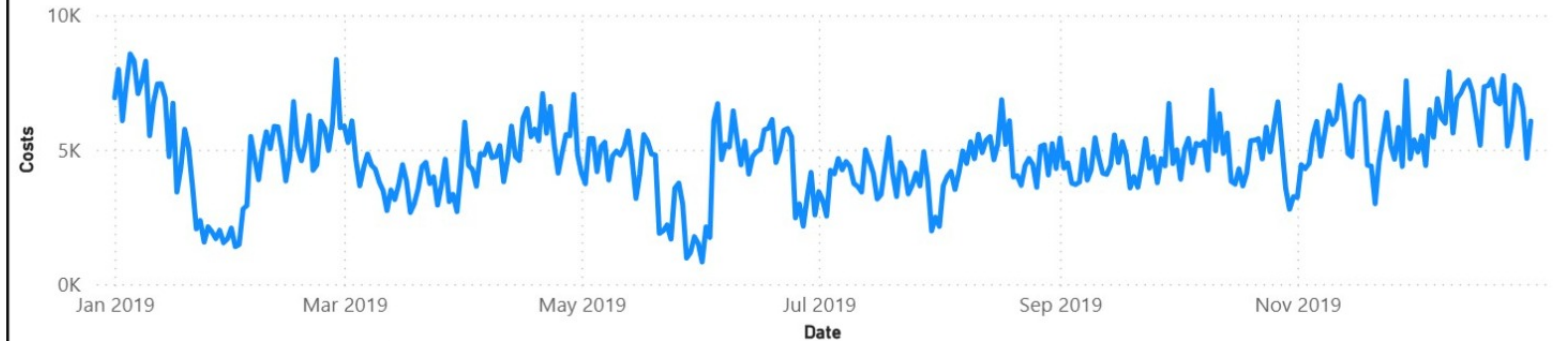
# 8. Marketing Correlation Analysis

## Marketing Costs

● Offline Marketing ● Online Marketing



## Marketing Costs by Date



## Executive Summary: Marketing Efficiency

**Data Distribution:** Shapiro-Wilk tests ( $p < 0.05$ ) and Q-Q plots confirm that data is **not normally distributed**. Therefore, **Spearman's rank correlation** was used for analysis.

**Budget Structure:** **Offline Marketing** dominates the budget at **59.87%** (\$1.04M), while **Online Marketing** accounts for **40.13%** (\$0.7M).

**Online Spend ( $\rho = -0.0488$ ):** Shows **no correlation** with the number of orders.

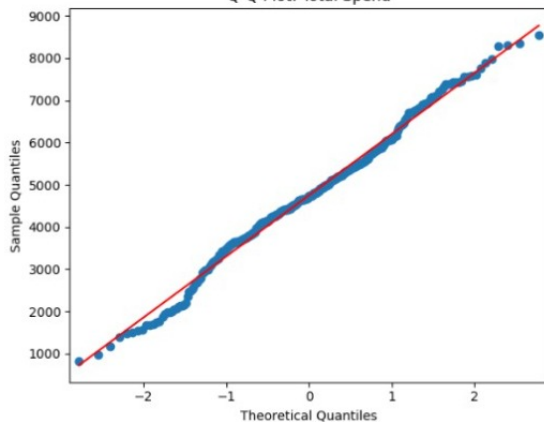
**Offline Spend ( $\rho = -0.0173$ ):** Shows **no correlation** with order volume. The vertical scatter plot proves that changing the offline budget has zero direct impact on daily orders.

**Total Spend ( $\rho = -0.0697$ ):** Shows **no correlation** with the number of orders.

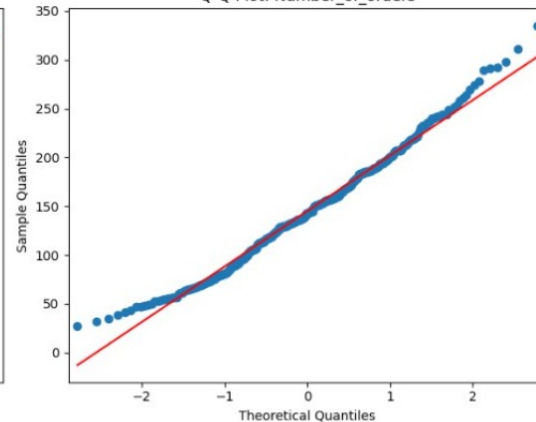
**Key Takeaway:** There is **zero monotonic relationship** between marketing investments (neither online nor offline) and the daily volume of orders.

**Business Insight:** Marketing budgets are currently ineffective at directly scaling order counts. Growth is driven by external factors not captured in this spending data.

Q-Q Plot: Total Spend



Q-Q Plot: Number\_of\_orders



Shapiro-Wilk Normality Test Results:  
Total\_Spend p-value: 0.02135  
--> NOT NORMAL  
Number\_of\_orders p-value: 0.00209  
--> NOT NORMAL

Spearman Rank Correlation  
(Target Variable: Number\_of\_orders)  
1. Total Spend : -0.0697  
2. Online Spend : -0.0488  
3. Offline Spend : -0.0373